

Answer **all** questions.

- 1 (a) Restriction mapping is a technique used to identify restriction enzyme cleavage sites. In an experiment, restriction mapping was used to investigate an unknown piece of DNA. This piece of DNA was 3 000 base pairs (bp) long.

The scientists first took plasmids that had one restriction site for the enzyme *KpnI* and one restriction site for the enzyme *BamHI*. The unknown piece of DNA was ligated into the multiple cloning site of the plasmid.

Fig. 1.1 shows the recombinant plasmid.

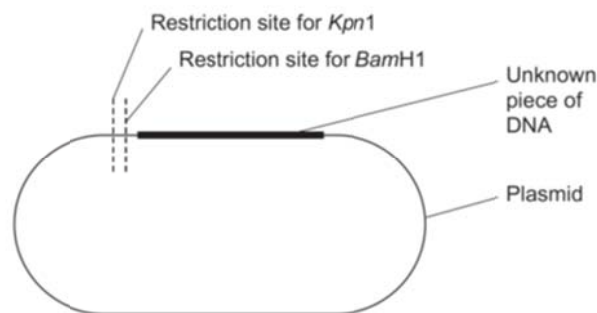


Fig. 1.1

When the scientists digested one of the recombinant plasmids with *KpnI*, they obtained two fragments. One fragment was measured as 1 000 bp. The other fragment was described as "very large".

- i. What does this show about the unknown piece of DNA? [2]

1. Has *KpnI* restriction site;
2. *KpnI* restriction site is unique/ cuts unknown piece only once;
3. 1000 bp from *KpnI* site of plasmid; (A: restriction site is found at the 1/3 position along the length of unknown DNA)

- ii. One of the fragments that the scientists obtained was described as "very large". What is represented by this very large fragment? [1]

(Most of) plasmid and rest of unknown DNA;

- iii. When the scientists digested another of the recombinant plasmids with *Bam*HI, they obtained three fragments. How many *Bam*HI restriction sites are there in the unknown piece of DNA? [1]

2;

- iv. When carrying out restriction mapping of the unknown DNA, scientists need to make sure that the restriction enzyme they have used has completely digested the DNA.

Suggest how they could use results from agarose gel electrophoresis to show that this DNA has **not** been completely digested? [1]

1. Restriction fragments add up to more than total length of original DNA;
2. Presence of undigested 3000bp fragment (+ some other smaller fragments)

- (b) The urate transporter is a crucial protein found primarily in the kidney. **Fig. 1.2** below shows how genetic engineering can be used to clone the human urate transporter gene in bacteria.

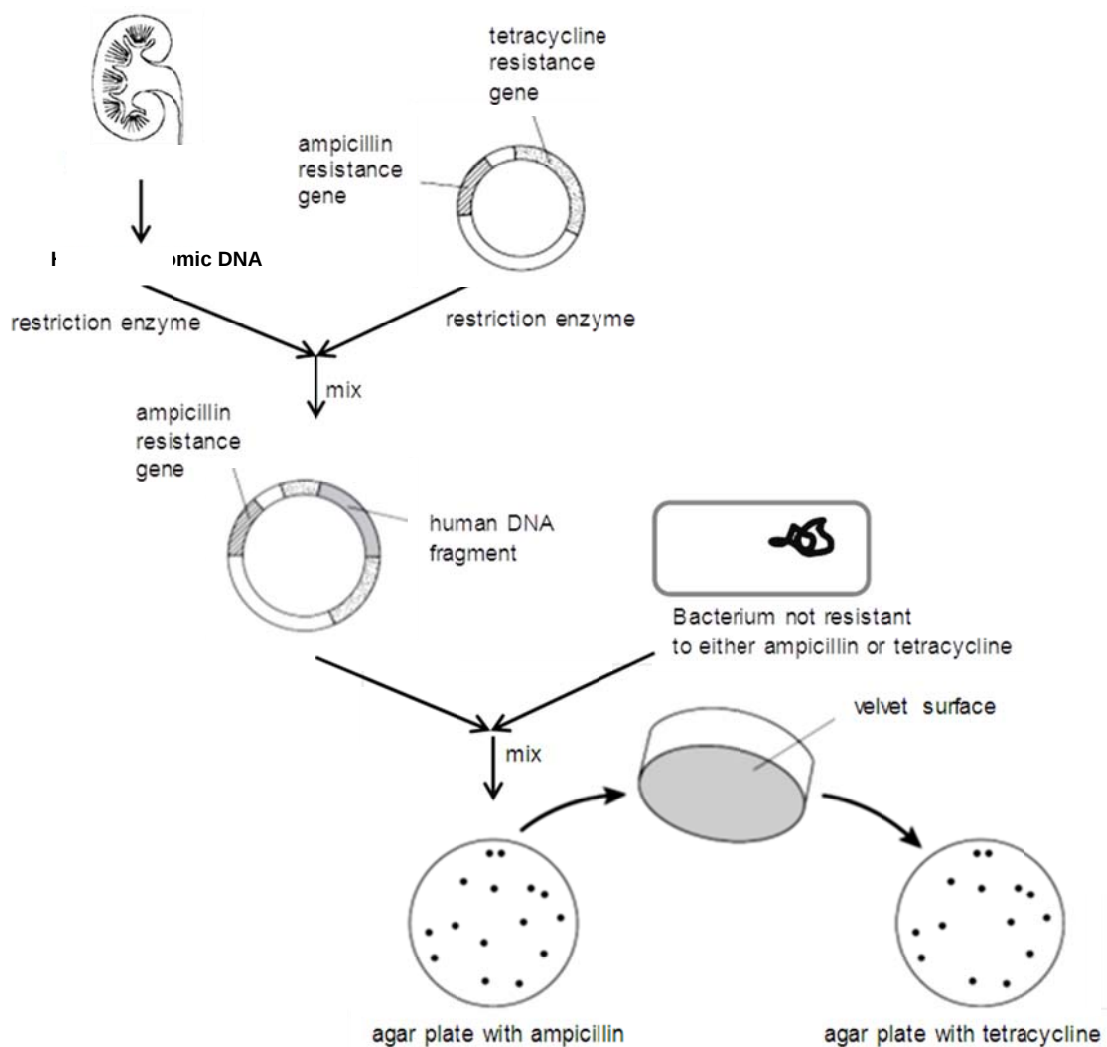


Fig. 1.2

- i. The human genomic DNA sample was digested with restriction enzyme. What is an important assumption made in this procedure to clone the urate transporter gene? [1]

assumes restriction enzyme doesn't cut within the urate transporter gene rendering it inactive;

- ii. Comment on making kidney the choice organ to extract the urate transporter gene in the procedure above. [2]

1. not necessary; (A: suggested use of other tissue)

2. urate transporter gene is present in all cells so can use any human tissue as starting material;
Or

3. procedure involves creating a genomic library so do not need to use a tissue with high levels of urate transporter mRNA;

- iii. The human DNA fragments are inserted into the gene coding for resistance to tetracycline. What effect will this have? [2]

1. gene coding for resistance to tetracycline no longer functional / bacteria not resistant to tetracycline/wtte;

2. insertional inactivation* has disrupted reading frame

- iv. During replica plating, why is the velvet surface used? [1]

So that bacteria stick to it / transfer of bacteria without damage

- v. Use an arrow (→) to point to the bacterial colony containing the human DNA fragment. [1]

Colony present on ampicillin plate but not on tetracycline plate

- vi. State one disadvantage in using the bacterium as a host cell for expressing the urate transporter gene. [1]

1. lack post-transcriptional modifications (source DNA / Gene of interest used cannot contain introns)

2. limited post-translational modifications (eukaryotic proteins may not be processed correctly)

[Total: 13]

- 2 Primordial dwarfism (PD) is a rare autosomal recessive disorder in which the lack of normal growth is not due to a deficiency of growth hormone. In linkage analysis using RFLPs, scientists can examine segments of DNA for a haplotype that is associated with PD. A haplotype refers to a group of gene loci or genetic markers that is inherited together from one parent on one chromosome.

Fig. 2.1 shows a pedigree of a family affected by PD followed by haplotype analysis with polymorphic microsatellite markers on a segment of chromosome 3 for individuals **II1** to **II5** and **III1**. Each microsatellite marker has 7 different known alleles.

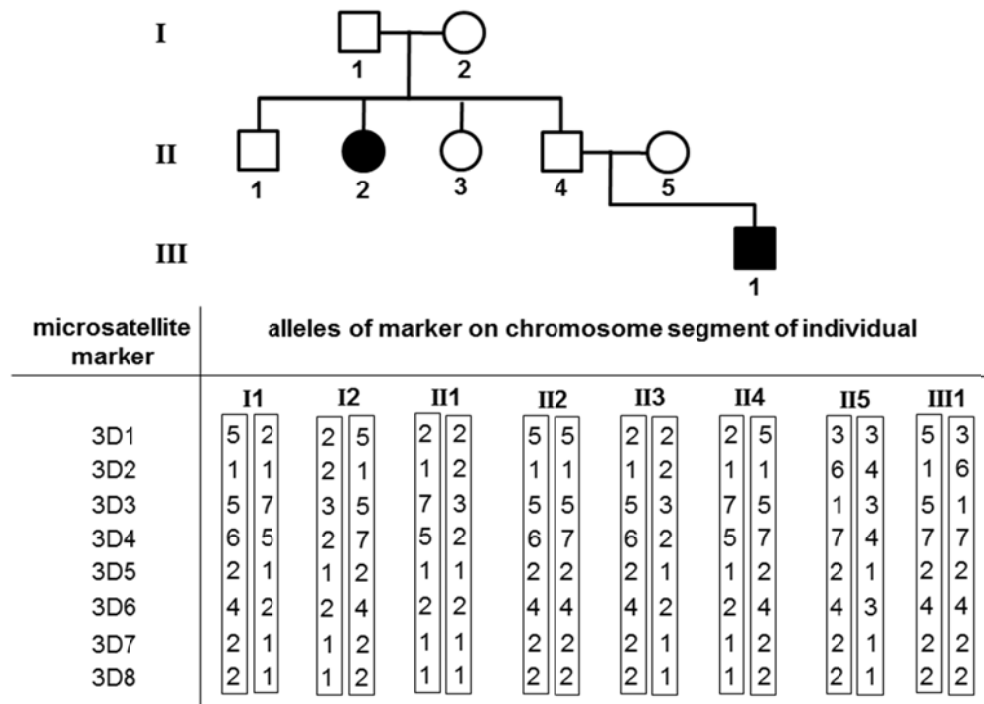


Fig. 2.1

(a) By examining the haplotypes in **Fig. 2.1**,

- i. identify all the microsatellite marker alleles that are associated with primordial dwarfism. [2]

1. 3D5 allele 2, 3D6 allele 4, 3D7 allele 2, 3D8 allele 2;

(1 mark for 2 correctly identified alleles)

- ii. identify which descendant is a recombinant. [1]

II3

- (b)i. One of the conditions for a microsatellite marker to be useful for disease detection is that it must be polymorphic.

Explain why the following 2 conditions are also necessary:

Condition 1: loci of the microsatellite marker allele must be tightly linked to the gene locus of the disease allele; [1]

1. So that the microsatellite marker allele and the disease allele always (higher chance) of co-segregating/(being inherited together);

OR

2. To minimise the chance of crossing over* at the region between the disease allele and marker allele so that the linkage of microsatellite marker allele to disease allele is not disrupted;

Condition 2: restriction sites must flank the microsatellite marker allele. [1]

1. Restriction enzyme should not cut within the microsatellite marker allele as it will result in the generation of the same sized RFLP fragments disallowing us to distinguish between the 7 different alleles;

OR

2. Ensures that excised RFLP fragments will reflect the variable lengths of each different allele (due to their difference in number of tandem repeats);

- ii. Suggest why the disease allele causing PD can only be mapped using linkage analysis. [1]

1. Nucleotide sequence of the disease allele is not known;
 2. Mutation in the disease allele in the genome is not known;
 3. The mutation in the disease allele did not introduce or remove a restriction site;
- R: There are no restriction sites in the disease allele – unlikely.

After more research, scientists discovered another RFLP marker that is also linked to PD. **Fig. 2.2** shows the two alleles of the RFLP marker with their restriction sites.

They carried out PCR to amplify the target DNA segment containing the RFLP allele. After which, restriction digestion using *BtgI* is done followed by gel electrophoresis to obtain the results shown in the PCR-RFLP analysis in **Fig. 2.3**.

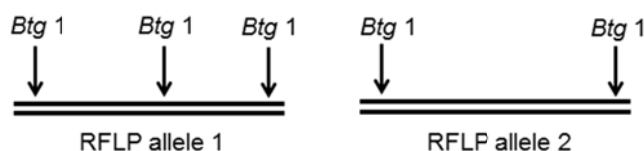


Fig. 2.2

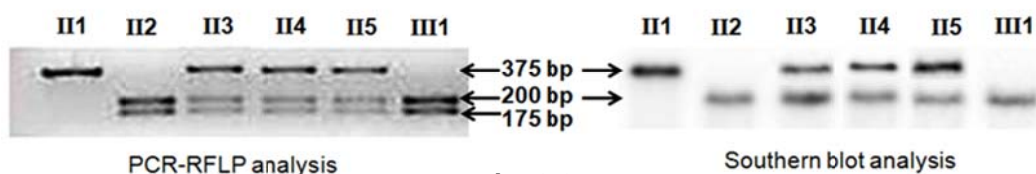


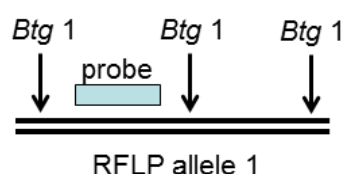
Fig. 2.3

(c) Outline the principle of gel electrophoresis. [3]

1. Negatively-charged DNA migrate out of well towards the positive electrode/anode when subjected to an electric field / current;
2. Meshwork of polymer fibers that makes up agarose gel impedes movement of DNA fragments;
3. thus longer DNA fragments migrate slower towards the positive electrode than shorter DNA fragments (or vice versa)

(d) Using information from **Fig. 2.2** and **Fig. 2.3**,

- i. indicate on the RFLP allele **1** in **Fig. 2.2** the position where the probe should hybridised to; [1]



- ii. explain the banding pattern of individual **II4** as seen in the PCR-RFLP analysis in **Fig. 2.3**. [2]

1. Using restriction enzyme *BtgI* for restriction digestion, RFLP allele 1 will be cut into two fragments of sizes 200bp and 175bp;
2. No restriction digestion within RFLP allele 2 thus resulting in (one fragment) 375bp band;
3. 3 bands shown for Individual II4 as he is heterozygous;

[Total: 12]

- 3 The *Syngonanthus mucugensis* species has significant economic value for the Brazilian ornamental dry flower business. However, extreme exploitation has rendered this species endangered.

Attempts at conventional propagation of this species by seeding have generated little success and studies are ongoing to develop alternative techniques that can assure propagation on a commercial scale. In a particular experiment, the potential of three different explants (stem, leaf and root) to form callus was studied using two plant growth regulators (BAP and NAA). The data from the experiment are shown in **Table 3.1** below.

Table 3.1

PGR (μM)		% of callus		
BAP	NAA	Stem	Leaf	Root
0	0	0	0	0
	1.34	0	0	0
	2.68	8.62	19.73	0
2.22	0	0	0	0
	1.34	0	0	0
	2.68	0	0	0
4.44	0	0	12.35	0
	1.34	0	0	0
	2.68	0	0	0

With reference to **Table 3.1**,

- (a)i. identify which explant seemed most suitable for use in micropropagation. [1]

leaf

- ii. explain your answer to (a) i. [4]

1. Root does not give rise to callus in the whole experiment.
2. Stem gives rise to 8.62% while leaves give rise to 19.73% (or 12.35%) callus (no need to mention concentration and type of growth regulator)
3. Leaves give rise to more callus compared to stem with data (either difference i.e. 11.11% or a factor of e.g. about 2.3x that of)
4. More callus giving rise to more plantlets

- iii. In a separate experiment, the ability of the explant to undergo organogenesis was studied using the same plant growth regulators. Data obtained is shown in **Table 3.2**.

Table 3.2

With reference to **Table 3.2**, explain which plant growth regulator has an inhibitory effect on shoot regeneration. [3]

1. NAA has inhibitory effect on shoot regeneration.
2. In the presence of BAP and absence of NAA, there is shoot regeneration.
3. however in the presence of both BAP and NAA, there is no shoot regeneration.
4. correct and relevant use of data (mention concentration of BAP and NAA and % shoot for both points 1 and 2)

To create transgenic maize which can produce the *Bacillus thuringiensis* toxin, the recombinant plasmid shown in **Fig. 3.1** was constructed. The recombinant plasmid also contains the gene for phosphomannose isomerase (PMI), an enzyme which allows plants cells to utilize mannose as a carbon source.

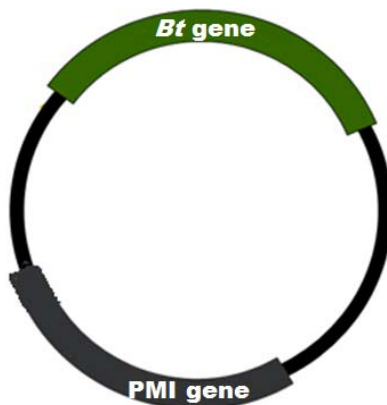


Fig. 3.1

To produce transgenic maize using the plasmid shown in **Fig. 3.1**, the procedure shown in **Fig. 3.2** was carried out.

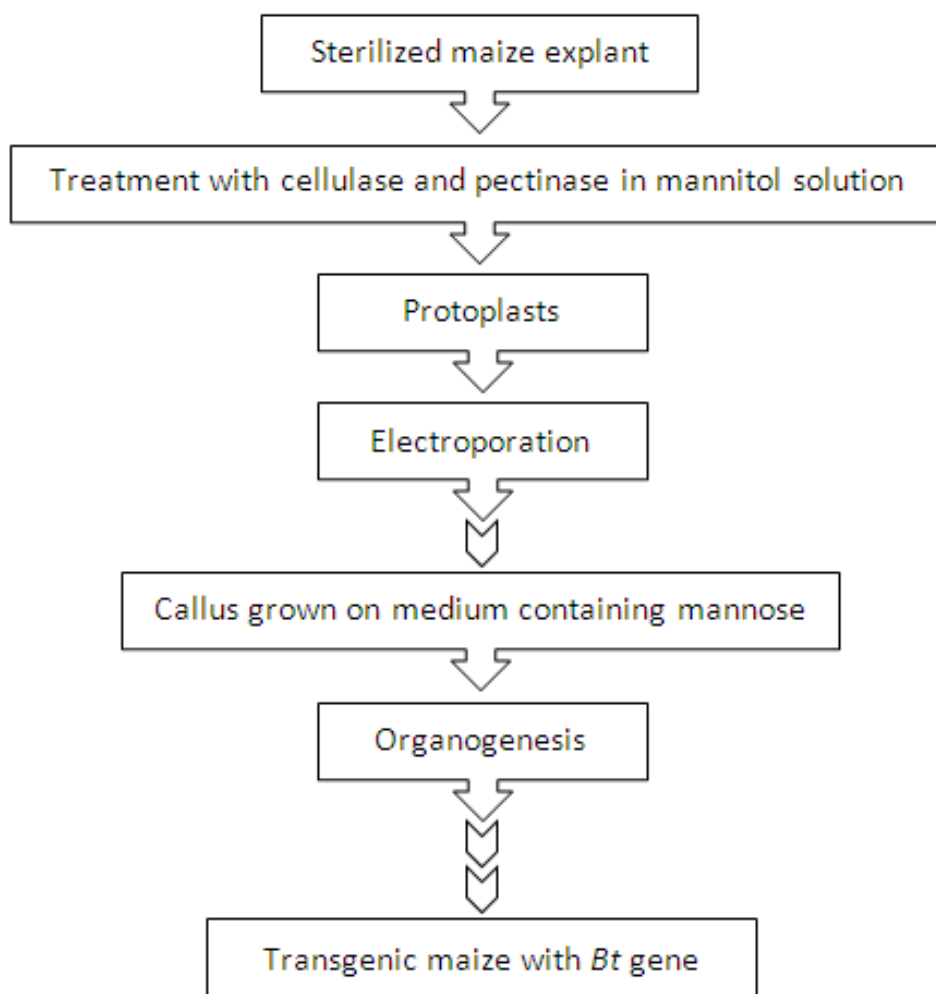


Fig. 3.2

With reference to **Fig. 3.2**,

(b)i. explain why the sterilized explant was treated with cellulase and pectinase before electroporation. [2]

1. To remove cell wall to form protoplasts;
2. Protoplasts which are lacking in cell wall can then take up plasmid containing *Bt* gene during electroporation / cell wall is a barrier to uptake of *Bt* gene during electroporation;

ii. In the preparation of protoplasts, mannitol must be present. In the absence of mannitol, no protoplasts are formed even if the sterilised explant was treated with cellulase and pectinase.

State what happens to the protoplasts if mannitol was not used. [1]

The protoplasts lysed;

- (c) Antibiotic or herbicide resistance genes are commonly included in the plasmid used for creating transgenic plants.

Explain why such genes were not needed in this case. [2]

- The PMI gene serves as the selectable marker* instead;
- In medium with only mannose as sole carbon source, transformed plant cells would have the PMI gene and survive OR non-transformed plant cells do not have the PMI gene and will not survive;

- (d) Explain the benefit of incorporating a *Bt* gene in the plasmid as shown in **Fig.3.1**. [2]

- Increase crop yield;
- Bt toxin is synthesized, making crops resistant to insects that consume it/plant is pest resistant or kill the pest that consume it;

[Total: 15]

Planning question

- 4 The purple colour of beet root (*Beta vulgaris*) is due to the water soluble betacyanin pigment molecules that are found in the vacuole of intact cells. Beet root pigment is used commercially as a food dye and heat is typically used as a method to extract the pigment from intact beet root cells.

A colorimeter can be used to measure the absorbance of light of 550 nm by betacyanin solution. The concentration of betacyanin is proportional to its absorbance. The colorimeter is shown in **Fig. 4.1** below.

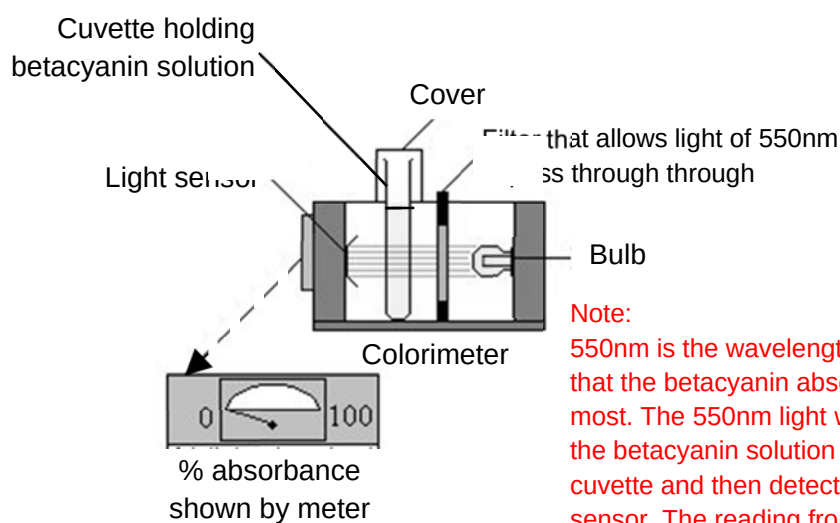


Fig. 4.1

Using the information and your own knowledge, design an experiment to investigate the effect of temperature on membrane permeability in beetroot.

Your plan must be based on the assumption that you have been provided with the following which you must use.

- Beet root
- Cork borer – *gives a cylinder (e.g 1 cm diameter) of beet root tissue*
- Scalpel – *for cutting.*
- Thermometer – *for checking the temperature of water bath*
- Source of boiling water and tap water – *to create water bath of a certain temperature and to add more for maintenance of temperature.*
- Distilled water – *for rinsing and immersing the beet root*
- Large 500 cm³ beaker – *used as a water bath*
- Colorimeter – *measures 550nm light absorbed by betacyanin pigment (absorbance) which is a reflection of the concentration of betacyanin that has leaked out of beet root*
- 2 cm³ cuvettes – *a small tube that holds solution whose absorbance is to be measured. Cuvette will be placed into colorimeter. 2cm³ is the maximum volume the cuvette can hold.*

You may also suggest other appropriate equipment to use.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by labeled diagrams, if necessary,
- identify independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results table(s) and graph(s) with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

1. Aim : To investigate the effect of temperature on membrane permeability in beetroot

2. Theory [T1 to T5 - max 3]:

(Main theory)

[T1] Betacyanin is found in beetroot cells is prevented from leaking out of cells by membranes (vacuole & plasma membrane) which are made up of phospholipid bilayer embedded with proteins

[T2] increasing temperature increases kinetic energy of phospholipids and membrane becomes more fluid or increasing temperature increases more intramolecular vibrations and membrane proteins are denatured

(Measurable quantity)

[T3, T4 and T5 - max 2]

[T3] the degree of permeability of the membrane will be reflected by the amount of betacyanin pigment leaking out of cells

[T4] The concentration of betacyanin is proportional to its absorbance measured by colorimeter.

(Predicted trend)

[T5] Increasing temperature increases the absorbance, which means permeability of the membrane is increased.

[DV] **Dependent variable**: % absorbance which indicates permeability of membrane [1]

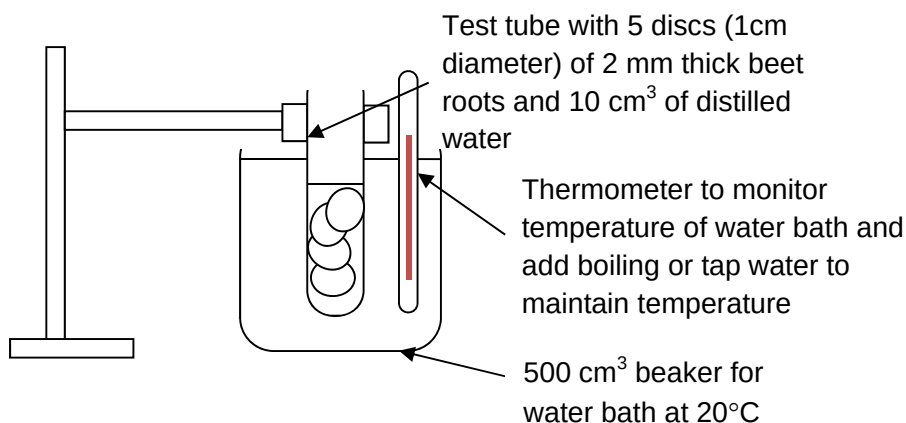
[IV] **Independent variable**: temperature 30, 40, 50, 60, 70°C [1]

3. **Procedure** (PAN CR)

a) [P] **Pilot test***

Conduct a pilot experiment to determine suitability of apparatus, suitability of range of independent variable (e.g. temperature range), optimum conditions, amount of materials used (number of pieces of beetroot, volume of distilled water) [1]

b) **Annotated diagram**



c) **Numbered steps**

Constant variables (What, How and Why) [max 3]

- (what) Surface area of beetroot – (how) Use similar sized beetroot sizes using cork borer and scalpel (2mm height, 1cm diameter discs) as (why) increasing surface area increases surface for pigment to leak out, thus affecting rate of diffusion of pigment.
- (what) Volume of distilled water in test tube – (how) Use 10 cm³ of distilled water as (why) different volume of water will affect the concentration of leaked pigment.
- (what) Duration of experiment – (how) 20 min as (why) the greater the duration, the longer the time where pigment may leak out.
- (what) Wavelengths of light used in the colorimeter when measuring absorbance – (how) Use 550 nm as (why) different wavelengths may be absorbed differently by the pigment, affecting accuracy of results.

[CV1] what and how variable 1;

[CV2] what and how variable 2;

[CVW] why for variables 1 and 2

1. Using a **cork borer*** (diameter 1cm), cut out cylinders of **beetroot***. Cut 75 pieces of **2 mm** thick discs from the cylinder using **scalpel***.
2. [B] Rinse the discs in running water until the water is **colourless**. (or idea of baseline for calibration of colorimeter) [1]
3. Place **5** pieces of washed beetroot discs in test tube. Prepare **2 more test tubes** with **beetroot discs** to serve as **replicates**, (why) **to check for anomalous results**.
4. Add **10 cm³** of **distilled water*** into 3 separate test tubes and place them in the **500 cm³ beaker*** with **30°C** water. Place the 3 test tubes with beetroot into the same water bath. Add **boiling or tap water***, where necessary, to **attain and maintain** the temperature. Use **thermometer*** to monitor temperature.
5. [E] Allow **2 min of equilibration time** for the (why) **temperature in the test tube to reach that of the water bath** at 30°C. [1]
6. Add the 5 beetroot discs to the 10 cm³ distilled water.
7. Incubate the 3 tubes in the 30°C water bath for **20 min**.
8. Decant the extract from tube and pour into liquid into a **cuvette***.
9. Fill a second cuvette with distilled water.
10. Place a filter that results in light of 550nm into the **colorimeter*** and use the distilled water to adjust the reading to zero absorbance. (do not mark for this)
11. Measure the % absorbance of solution using colorimeter.
12. Repeat steps 1-10 for 40°C, 50°C, 60°C, 70°C.
13. [RR] Repeat the entire experiment **twice** to **check for reproducibility**. [1]

d) [C] **Control** [1]

Place 5 washed beetroot disc into 10 cm³ of distilled water into a test tube place on test-tube rack at room temperature. All other conditions remain the **same**. This shows that raising temperature will result in increased permeability of membrane.

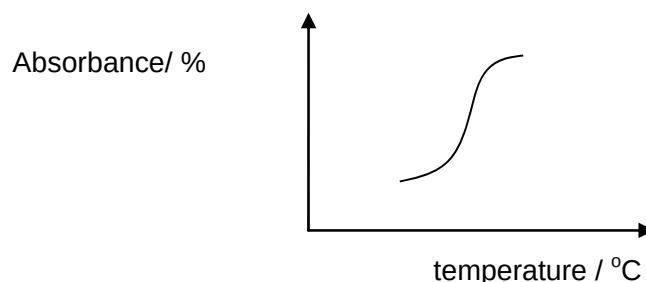
4. **Data** Recording and Processing

Record and process data as follows:

[T] **Table showing intensity of colour of extract** [1]

Temperature / °C	Absorbance/%			
	Replicate 1	Replicate 2	Replicate 3	Average
30				
40				
50				
60				
70				

[G] Graph showing intensity of colour of extract obtained from incubation at different temperature [1]



Note:

- trend accepted as long as there is increasing absorbance with increasing temp

5. [RP] Risks & Precautions [1]

- When using scalpel, cut the beetroot away from the hand to avoid getting cut.
- Use insulating gloves when handling hot water to prevent scalding
- Dry your hands before switching the colorimeter on or off to avoid being electrocuted.

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labeled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections (a), (b) etc, as indicated in the question.

5 (a) Explain the normal functions of stem cells in a living organism. [6]

- Stem cells are a group of undifferentiated* and unspecialized* cells ;
- Ability to differentiate* into specialized cells when appropriate molecular signals are present
- Zygotic stem cells are totipotent* and can differentiate into all cell types (including extra-embryonic tissue) / make up the whole organism ;
- Embryonic stem cells and pluripotent* which can differentiate into all cell types except the extra-embryonic tissue/placenta ;
- adult stem cells are multipotent* which can differentiate into limited range of related cell types ;
- any example of adult stem cells i.e. haematopoietic stem cells which differentiate into red and white blood cells or neural stem cells from our central nervous system can proliferate and differentiate into both neurons and supporting cells to replace those that died etc ;
- Stem cells are capable of self renewal* to ensure a constant pool of stem cells with the same development potential ;
- Stem cells divide by mitosis* to produce genetically identical daughter cells ;
- function: for growth and development ;
- function: to regenerate or replace cells lost due to cell death and injury ;

(b) Discuss the social and ethical concerns for the use of gene therapy. [6]

1. Gene therapy is a method of treating genetic diseases which involves the introduction of a normal functional allele into cells of an affected individual where it will express the normal protein ;
2. Gene therapy which involves an alteration to the genes of somatic cells affects only that individual, but an alteration to the genes of germline cells affect offspring and successive generations ;
3. Social issue: injustice and unequal resource allocation as gene therapy is very expensive. There are concerns that this treatment is being made available only to those who can afford it ;
4. Ethical issue: There is concern that many gene therapy candidates are children who are too young to understand the ramifications of gene therapy treatment and can become unwitting participants ;
5. Ethical issue: Germ-line gene therapy involves research on early embryos which may be considered by some as living and death of embryos in the process of gene therapy (e.g. extraction of inner cell mass) is unethical ;
6. Ethical/social issue: Abuse of gene therapy for genetic enhancement could exacerbate problems of social discrimination, social inequality and a lowered tolerance for human diversity ;

(c) Describe cystic fibrosis and how it can be treated using non-viral gene delivery system. [8]**Cystic fibrosis description (max 4)**

1. Cystic fibrosis (CF) is a autosomal recessive disease;
 2. caused by mutation of cystic fibrosis transmembrane conductance regulator* (CFTR/ CFTCR) gene;
 3. The mutation of CFTR gene in chromosome 7 is a deletion of 3 bases resulting in the loss of phenylalanine in the polypeptide;
 4. A normal CFTR gene will code for a protein that function as a gated channel protein to facilitate the transport of chloride ions;
 5. as a result of this loss of function* mutation, channels no longer able to transport chloride ions out of the epithelial cell (as in the lung);
 6. causing thick mucus in lungs, resulting in blockage/congestion of airway in the lung that reduces gaseous exchange leading to breathing difficulties
- Or
- pancreatic duct is choked by thick mucus preventing release of enzymes resulting in indigestion ;
7. thick mucus provides suitable environment for bacteria growth which makes patient susceptible to chronic lung infection resulting in death (lowering of life expectancy, not living beyond 30years old);

Treatment (max 4)

8. to treat CF, a normal functioning allele* of CFTR is isolated and inserted into a plasmid vector through recombinant techniques;
9. this DNA construct is then packaged in the aqueous core of the microscopic lipid envelopes/phospholipid bilayer sphere forming liposomes* ;
10. delivered to epithelial cell in the lung by aerosol spray (nasal spray);
11. the liposome fuses with the plasma membrane of the epithelial cell lining the lung and releases the DNA into the cell (this process is called lipofection);
12. the plasmid vector enters nucleus and the normal CFTR allele is transcribed and translated;
13. to produce normal CFTR protein which can be incorporated into plasma membrane to facilitate the transport of chloride;

[Total: 20]